Logic-Based Vending Machine Controller

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- **Selected Part Description**

Selected Component: Coin Evaluation and Dispense/Change Logic Circuit

The chosen decision-making component is the core logic circuit of the vending machine. This component receives three binary inputs (A, B, and C) representing inserted coins (A=5, B=5, C=10). It constantly evaluates these inputs to make a real-time decision on whether the inserted amount meets the required product price of 15. Based on this evaluation, the circuit decides to activate the **Out** signal to dispense the product and/or the **Change** signal to return excess money (5) if the total inserted is 20.

- **Truth Table**

The truth table evaluates all 8 possible combinations of the 3 input coins.

Inputs: A (5 coin), B (5 coin), C (10 coin)

Outputs: Out (Dispense product), Change (Return 5)

A (5 coin)	B (5 coin)	C (10 coin)	Out (Dispense)	Change (Return 5coin)
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	**1**	0
1	0	0	0	0
1	0	1	**1**	0
1	1	0	0	0
1	1	1	**1**	**1**

- **Karnaugh Maps (K-Maps) and Simplification Steps**

A. K-Map for 'Out' Signal

Out = Σm(m3,m5,m7)

A \ BC	B'C'	BC'	BC	BC'
A'	0	0	**1**	0
A	0	**1**	**1**	0

B. K-Map for 'Change' Signal

Change = Σm (m7)

A \ BC	B'C'	B'C	BC	BC'
A'	0	0	0	0
A	0	0	**1**	0

- **Boolean Equations**

The final simplified Boolean expressions derived from the K-Maps are:

Equation for dispensing the product:

```
Out  =  C.(A + B)
```

Equation for returning change:

```
Change  =  A · B · C
```